Diagnosing and Mitigating Object Hallucination in Vision-Language Models via Parameter-Efficient Low-Rank Adaptation

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Abstract

Large Vision-Language Models (VLMs) can generate affirmative claims about objects that are absent from an image. This phenomenon, generally referred to as object hallucination, is associated with language priors, object co-occurrence statistics, weak visual grounding, and generation-time confirmation bias. This work investigates object hallucination using the Polling-based Object Probing Evaluation (POPE) benchmark and studies whether a lightweight parameter-efficient adaptation strategy can increase reliance on image evidence under severe hardware constraints.

The experimental system uses `HuggingFaceTB/SmolVLM-500M-Instruct`, a 509.7M-parameter multimodal model evaluated on adversarial POPE probes. Three configurations are examined: standard zero-shot prompting, explicit visual-verification prompting, and Low-Rank Adaptation (LoRA). The final LoRA configuration modifies only the attention `q_proj` and `v_proj` modules with rank $r = 16$ and scaling factor $\alpha = 32$. Under a 6 GB VRAM limit, gradient checkpointing and PEFT-specific state-dictionary handling are required to make training and checkpoint restoration reliable.

The principal mitigation reduces the False Positive Ratio (FPR) from 0.20 to 0.16, corresponding to a 20% relative reduction. Only 2,228,224 parameters, or 0.4372% of the total model parameters, are trainable. Overall accuracy remains 0.81, while the Yes/No response ratio changes from 0.51 to 0.47. A critical diagnostic finding is that the training data must be strictly balanced at 50/50: an uncurated training subset caused affirmative class collapse, increasing adversarial FPR to 0.42 and Yes/No bias to 0.70. The study therefore demonstrates both a practical hallucination-mitigation mechanism and the importance of controlled dataset composition in evaluating multimodal behavior.

Index Terms

Vision-Language Models, Object Hallucination, POPE, LoRA, Parameter-Efficient Fine-Tuning, Visual Grounding, Multimodal Learning

I. Introduction

Vision-Language Models combine visual representations with autoregressive language models to perform multimodal reasoning, visual question answering, image understanding, and instruction following. Although these systems can achieve strong performance across a wide range of multimodal tasks, they may generate statements that are linguistically plausible but unsupported by the visual input.

Object hallucination is a particularly important instance of this problem. A VLM may answer "yes" to a query about an object that does not actually occur in the image. Such behavior can arise because the language model has learned strong statistical associations between objects, scenes, and textual descriptions. Consequently, the model may generate a semantically likely answer even when the corresponding visual evidence is weak or absent.

The POPE benchmark was designed specifically to diagnose this type of behavior by converting object hallucination into a controlled binary classification problem [1]. Rather than evaluating only free-form generation, POPE asks explicit object-presence questions and measures the resulting confusion between present and absent objects.

The central research question of this work is:

Can a lightweight parameter-efficient adaptation strategy make a VLM rely more on image evidence than language and object co-occurrence priors, thereby reducing false-positive object claims on POPE without sacrificing balanced accuracy?

Three configurations are studied:

1) **Baseline zero-shot probing**, using the model's standard conversational prompt;
2) **Visual-verification prompting**, which explicitly instructs the model to depend on visible image evidence and answer "no" when the object cannot be clearly verified;
3) **LoRA-based adaptation**, which updates a small number of attention parameters while keeping the pretrained backbone frozen.

The work is intentionally conducted under a strict local hardware constraint. The available accelerator is an NVIDIA RTX 4050 Laptop GPU with approximately 6 GB of VRAM. This constraint motivates a parameter-efficient training strategy rather than full-model fine-tuning.

II. Object Hallucination and POPE Evaluation

A. POPE Problem Formulation

The Polling-based Object Probing Evaluation (POPE) models object hallucination as binary hypothesis testing. Given an image I and a target object o, the model receives a query of the form:

```
Is there a <object> in the image? Answer yes or no.
```

The model produces

$$\hat{y} \in \{\text{yes}, \text{no}\}.$$

POPE constructs negative object queries according to different sampling distributions. The three principal settings are:

- **Random:** absent objects are selected randomly from non-present categories.
- **Popular:** absent objects are sampled using global object-frequency statistics.
- **Adversarial:** absent objects are chosen according to strong co-occurrence relationships with objects actually present in the scene.

The adversarial setting is the primary diagnostic environment used in this study because it directly stresses learned language and co-occurrence priors.

B. Hallucination Mechanisms

The observed hallucination behavior can be interpreted through several interacting mechanisms.

Language priors: the causal language backbone assigns high probability to tokens that are statistically plausible even when visual evidence is insufficient.

Object co-occurrence bias: the model can infer an object because it frequently occurs together with a visible object. A visually detected knife, for example, can create pressure toward predicting a fork even when no fork is visible.

Weak visual grounding: visual features are projected into the language-model representation space, and language attention patterns can dominate when visual evidence is compressed or ambiguous.

Generation pressure: instruction-following language models can exhibit a tendency toward affirmative confirmation, which manifests as a measurable Yes/No response bias.

These mechanisms motivate evaluating hallucination using more than a single accuracy value.

III. Evaluation Metrics

Let TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives.

A. Overall Accuracy

Overall accuracy measures the fraction of all POPE queries that are classified correctly:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{1}$$

B. False Positive Ratio

The primary hallucination metric is the False Positive Ratio:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \tag{2}$$

A false positive corresponds directly to an hallucinated claim that an absent object is present. Lower FPR therefore indicates better resistance to object hallucination.

C. Yes/No Bias

The affirmative response ratio is defined as

$$\text{Yes Ratio} = \frac{\text{TP} + \text{FP}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{3}$$

For a balanced evaluation set, a value close to 0.50 indicates that the system is not strongly biased toward affirmative or negative responses. However, very low Yes Ratios may also indicate pathological negative collapse. Consequently, Yes/No Bias must be interpreted together with FPR and accuracy.

D. Precision, Recall, and F_1

Precision is

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{4}$$

Recall is

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{5}$$

and the harmonic mean is

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{6}$$

An effective mitigation should reduce hallucinated positives without simply forcing the model toward “no” on every query.

IV. Experimental System

A. Hardware and Runtime Environment

All experiments were performed locally on an NVIDIA GeForce RTX 4050 Laptop GPU. The relevant environment is:

- GPU: NVIDIA GeForce RTX 4050 Laptop GPU;
- VRAM: approximately 6,141 MiB ($\sim$ 6 GB);
- CUDA: 13.0;
- PyTorch: 2.14.0;
- Host operating system: Fedora Linux, x86-64;
- Numerical precision: native `bfloat16`.

The 6 GB ceiling strongly constrains multimodal backpropagation because activation tensors, gradients, and optimizer states can substantially exceed the memory required for the model weights alone.

B. VLM Backbone

The selected model is:

`HuggingFaceTB/SmolVLM-500M-Instruct`

The model contains approximately 509,710,528 parameters. It was selected because its sub-billion-parameter scale makes local experimentation feasible while retaining a sufficiently expressive multimodal architecture for diagnosing visual grounding failures.

The model was loaded in native `bfloat16` precision and occupied approximately 1.2 GB of VRAM in its idle configuration.

V. Methodology

A. Intervention 1: Zero-Shot Baseline

The baseline uses the standard processor chat template:

```
USER: <image>
{question}
Answer yes or no.
ASSISTANT:
```

Generation uses greedy decoding with a maximum of five new tokens. The generated response is isolated from prompt tokens and normalized into a strict binary “yes” or “no” prediction.

This baseline establishes the model’s natural hallucination behavior without parameter or instruction intervention.

B. Intervention 2: Visual-Verification Prompting

The second configuration adds an explicit instruction intended to suppress affirmative confirmation when visual evidence is insufficient:

```
USER: <image>
{question}
Look only at the image.
If the object is not clearly visible,
```

```
you must say no. Answer yes or no.
ASSISTANT:
```

The underlying hypothesis is that explicit negative-evidence instructions can shift generation toward visual verification. The intervention requires no model updates, but it increases prompt length and changes the inference-time computational cost.

C. Intervention 3: LoRA Adaptation

The final mitigation uses Low-Rank Adaptation (LoRA), introduced by Hu et al. [2]. Instead of optimizing the complete pretrained model, LoRA freezes a weight matrix W_0 and introduces a low-rank update:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA, \tag{7}$$

where

$$B \in \mathbb{R}^{d \times r}, \qquad A \in \mathbb{R}^{r \times k}, \qquad r \ll \min(d, k).$$

The LoRA adapters are applied specifically to the attention query and value projections:

$$\mathcal{M}_{\text{LoRA}} = \{\texttt{q_proj}, \texttt{v_proj}\}.$$

The final configuration uses

$$r = 16, \qquad \alpha = 32.$$

Only 2,228,224 parameters are trainable. Relative to the full model,

$$\frac{2,228,224}{509,710,528} \times 100 = 0.4372\%. \tag{8}$$

Thus, more than 99.56% of the pretrained model remains frozen.

D. Masked Causal Language Modeling Objective

Training uses the standard causal language-modeling objective, but only response tokens contribute to the loss:

$$\mathcal{L}(\theta) = -\sum_{t=L_{\text{prompt}}+1}^{L_{\text{total}}} \log P_\theta \left(y_t \mid y_{<t}, X_v, X_q \right). \tag{9}$$

Labels corresponding to prompt positions

$$t \leq L_{\text{prompt}}$$

are assigned the ignore value -100. This prevents the model from being penalized for reproducing the input prompt and concentrates optimization on the desired “yes” or “no” response.

E. Optimization Configuration

The training configuration is summarized below:

TABLE I
LoRA TRAINING CONFIGURATION.

Parameter	Configuration
Model	`SmolVLM-500M-Instruct`
Target modules	`q_proj`, `v_proj`
LoRA rank	$r = 16$
LoRA scaling	$\alpha = 32$
Trainable parameters	2,228,224
Trainable parameter ratio	0.4372%
Optimizer	AdamW
Learning rate	2×10^{-4}
Batch size	1
Gradient accumulation	1
Precision	`bfloat16`
Training category	POPE Random
Evaluation category	POPE Adversarial

VI. Dataset Curation and Anti-Leakage Strategy

The evaluation split consists of 100 adversarial POPE samples. To prevent evaluation contamination, the training pool is sampled exclusively from the random category rather than the adversarial evaluation category.

The final training pool contains:

$$100 \text{ positive samples} + 100 \text{ negative samples} = 200 \text{ training samples}.$$

The positive and negative examples are concatenated and shuffled using seed 42.

A. Why Strict 50/50 Balance Is Necessary

The experiments revealed that class balance is not a minor preprocessing detail. It directly determines whether the model learns visual verification or simply exploits the class distribution.

During an initial experiment, LoRA was trained on an uncurated 200-sample slice. The resulting adapter showed a severe affirmative class collapse:

$$\text{FPR} : 0.16 \rightarrow 0.42,$$

$$\text{Yes Ratio} : 0.47 \rightarrow 0.70,$$

and

$$\text{Accuracy} : 0.81 \rightarrow 0.78.$$

The model could reduce training loss by predicting the majority class rather than learning a better mapping from image evidence to binary object presence.

The final methodology therefore enforces an exact 50/50 class distribution. This constraint prevents the adapter from using answer-frequency statistics as a shortcut.

The experiment establishes the following diagnostic principle:

Hallucination mitigation must be evaluated under controlled class balance; otherwise, improvements or degradations in FPR can be caused by response-distribution collapse rather than improved visual grounding.

VII. VRAM Optimization and PEFT Engineering

A. Gradient Checkpointing

The 6 GB VRAM limit makes conventional multimodal backpropagation difficult. Gradient checkpointing is therefore enabled using:

```
peft_model.gradient_checkpointing_enable()
```

Checkpointing trades additional computation for lower activation memory. Instead of retaining every intermediate activation from the forward pass, selected activations are recomputed during the backward pass.

For compatibility with PEFT and checkpointed computation, input gradients are enabled through:

```
peft_model.enable_input_require_grads()
```

This allows adapter training to remain within the available VRAM budget.

B. Explicit CUDA Memory Management

The training and inference pipeline additionally invokes:

```
gc.collect()
torch.cuda.empty_cache()
```

before major memory-intensive operations. These calls are used to reduce avoidable memory retention and allocator fragmentation.

C. PEFT State-Dictionary Bug Fix

A significant engineering problem appeared when checkpointing the model through the native PyTorch call:

```
model.state_dict()
```

In PEFT and quantized environments, a conventional state dictionary can contain internal quantization-related entries such as:

```
.absmax
.quant_map
.quant_state.bitsandbytes__nf4
```

Restoring such a dictionary through ordinary model-state loading can result in namespace and key-mismatch collisions.

The final checkpointing protocol isolates only the adapter state using:

```
get_peft_model_state_dict(peft_model)
```

and restores the adapter through:

```
set_peft_model_state_dict(peft_model, weights)
```

This adapter-only state management avoids unnecessarily serializing the frozen backbone and prevents quantization-state collisions during checkpoint restoration.

VIII. Experimental Results

A. Comparative POPE Performance

The three configurations were evaluated on the 100-sample POPE adversarial evaluation set.

TABLE II
POPE adversarial probing performance. Lower FPR is desirable; accuracy is retained as a balanced diagnostic metric.

Metric	**Baseline**	**Prompting**	**LoRA (Ours)**
Overall Accuracy	0.8100	0.8200	**0.8100**
False Positive Ratio (FPR)	0.2000	0.1800	**0.1600**
Yes/No Ratio	0.5100	0.5000	0.4700
Precision	0.8039	0.8200	**0.8298**
Recall	0.8200	0.8200	0.7800
F_1	0.8119	0.8200	0.8041
Relative FPR Reduction	Baseline	10.0%	**20.0%**
Trainable Parameters	0	0	**2,228,224**
Trainable Ratio	0.0%	0.0%	**0.4372%**
Inference Throughput	3.65 it/s	1.27 it/s	1.21 it/s

Note: Precision, Recall, and F_1 values are derived from the reported Accuracy, FPR, and Yes/No Ratio under the balanced 50/50, 100-sample evaluation protocol.

B. False Positive Ratio Reduction

The baseline produces

$$\mathrm{FPR}_{\mathrm{baseline}} = 0.20.$$

Visual-verification prompting reduces this to

$$\mathrm{FPR}_{\mathrm{prompt}} = 0.18.$$

The LoRA configuration further reduces it to

$$\boxed{\mathrm{FPR}_{\mathrm{LoRA}} = 0.16}.$$

The relative reduction from the baseline is

$$\frac{0.20 - 0.16}{0.20} \times 100 = 20.0\%. \tag{10}$$

Thus, the principal mitigation achieves a 20% relative reduction in false-positive object claims.

C. Confusion-Matrix Interpretation

Because the evaluation set contains 100 balanced examples, the reported aggregate values correspond to the following integer confusion matrices.

For the baseline:

$$\mathrm{TP} = 41, \qquad \mathrm{TN} = 40, \qquad \mathrm{FP} = 10, \qquad \mathrm{FN} = 9.$$

Therefore,

$$\mathrm{Precision} = \frac{41}{41 + 10} = 0.8039,$$

$$\mathrm{Recall} = \frac{41}{41 + 9} = 0.8200,$$

and

$$F_1 \approx 0.8119.$$

For the prompt-engineered model:

$$\mathrm{TP} = 41, \qquad \mathrm{TN} = 41, \qquad \mathrm{FP} = 9, \qquad \mathrm{FN} = 9,$$

which gives

$$\mathrm{Precision} = 0.8200, \qquad \mathrm{Recall} = 0.8200, \qquad F_1 = 0.8200.$$

For the LoRA model:

$$\mathrm{TP} = 39, \qquad \mathrm{TN} = 42, \qquad \mathrm{FP} = 8, \qquad \mathrm{FN} = 11.$$

Hence,

$$\mathrm{Precision} = \frac{39}{39 + 8} = 0.8298,$$

$$\mathrm{Recall} = \frac{39}{39 + 11} = 0.7800,$$

and

$$F_1 \approx 0.8041.$$

The resulting pattern indicates a shift toward reducing false-positive predictions. Precision increases while Recall decreases modestly.

D. Yes/No Bias

The baseline affirmative-response ratio is 0.51, effectively centered around the balanced 0.50 operating point.

Prompt engineering produces a ratio of exactly 0.50.

The LoRA model produces a ratio of 0.47.

This shift is consistent with reduced affirmative false positives. Importantly, it is substantially different from the failed class-collapse experiment, in which the Yes/No ratio increased to approximately 0.70.

IX. Training Dynamics

The balanced LoRA configuration produced stable optimization before early stopping. The recorded loss trajectory is shown in Table III.

TABLE III
LoRA training dynamics.

Step	Loss	State / Action
0	7.2730	Initial forward pass and adapter initialization
20	1.2473	Rapid parameter alignment
400	0.0862	Stable convergence
600	0.0766	Reduction of false-confirmation logits
800	0.0026	Global minimum; best checkpoint saved
1000	0.0048	Generalization plateau
1200	0.1440	Patience limit reached; early stopping triggered

The best observed training loss was 0.0026. Once the loss ceased improving and subsequently increased, early stopping was triggered and the best adapter state was restored through the PEFT-specific checkpointing protocol.

X. Diagnostic Failure Analysis and Ablations

A. Class-Collapse Failure

The most important failure mode was observed when the training set was not explicitly balanced.

Under an uncurated 200-sample training slice, the adapter learned a strong affirmative prior. The resulting adversarial metrics deteriorated to:

$$\mathrm{FPR} = 0.42, \qquad \text{Yes Ratio} = 0.70, \qquad \mathrm{Accuracy} = 0.78.$$

This behavior is characteristic of class collapse: the model discovers that predicting the majority response reduces causal loss more easily than learning a reliable visual grounding function.

The successful remediation was strict 50/50 balancing:

$$100 \text{ yes} \quad + \quad 100 \text{ no}.$$

This balance forces the adapter to learn distinctions between visual presence and absence rather than exploiting class-frequency shortcuts.

B. Prompting Ceiling

The prompt-based intervention provides an immediate reduction in FPR:

$$0.20 \rightarrow 0.18.$$

However, throughput falls from

$$3.65 \text{ it/s}$$

to

$$1.27 \text{ it/s}.$$

The longer instruction increases token-processing cost and does not modify the underlying pretrained attention projections. Prompting can therefore change the generation policy without directly changing the learned multimodal representation pathway.

C. LoRA Versus Instruction-Only Intervention

LoRA produces

$$\text{FPR} = 0.16,$$

while preserving overall accuracy at

$$\text{Accuracy} = 0.81.$$

The intervention updates only 0.4372% of the total parameters. This provides a direct mechanism for changing how attention layers process multimodal information without the computational and memory requirements of full-model fine-tuning.

D. State-Dictionary Collision

The PEFT state-management failure highlights a second category of engineering risk. The semantic performance of the model can be correct while the training pipeline remains unreliable if checkpoints are serialized incorrectly.

Using:

```
get_peft_model_state_dict()
```

for saving and

```
set_peft_model_state_dict()
```

for restoration isolates the trainable adapter weights and avoids incompatible quantization-related entries.

This is particularly relevant under constrained environments, where quantization and PEFT are often combined to fit multimodal models into limited GPU memory.

XI. Alternative Hallucination Mitigation Strategies

Visual Contrastive Decoding (VCD) provides an alternative inference-time mitigation strategy [4]. VCD contrasts the generation behavior under the original image with generation behavior under a visually distorted input. The objective is to suppress output components that persist independently of reliable visual evidence.

The key distinction between the studied approaches is:

- **Prompt engineering** changes the textual instruction but does not modify model parameters;
- **LoRA** modifies a small set of trainable attention parameters while keeping the pretrained model frozen;
- **VCD** changes the decoding process itself and requires additional inference computation rather than adapter training.

The present implementation selects LoRA because the objective is to obtain a persistent lightweight adaptation while respecting the 6 GB VRAM constraint.

XII. Broader Benchmark Context

POPE is particularly useful for controlled object-hallucination diagnosis because it transforms hallucination into a binary object-presence decision. Nevertheless, it does not capture every form of hallucination that may appear during open-ended multimodal generation.

THRONE evaluates object hallucination in free-form VLM generations and therefore complements the binary POPE protocol [5].

A-OKVQA addresses visual question answering that requires external world knowledge and provides a complementary evaluation of multimodal reasoning [6].

LLaVA established the broader visual-instruction-tuning paradigm for conversational multimodal systems and provides an important architectural foundation for modern VLMs [3]. More comprehensive surveys of VLM hallucination identify interacting causes including language priors, weak grounding, and generation-induced behavior [7].

These resources suggest that future evaluation should combine binary object probing with free-form generation and knowledge-intensive visual reasoning rather than relying on a single benchmark.

XIII. Discussion

The experimental results demonstrate several important characteristics of hallucination mitigation.

First, the principal improvement is obtained through a very small parameter update. LoRA trains only 2.23 million parameters, corresponding to 0.4372% of the model, while reducing adversarial FPR by 20% relative to the zero-shot baseline.

Second, the improvement is not equivalent to simply increasing the frequency of "no" responses. The Yes/No ratio moves from 0.51 to 0.47, while accuracy remains at 0.81. This is materially different from the failed training condition, in which the Yes/No ratio reached 0.70 and overall accuracy dropped to 0.78.

Third, the reduction in FPR is accompanied by a Precision-Recall trade-off. Precision improves from approximately 0.804 to 0.830, whereas Recall decreases from 0.82 to 0.78. This behavior is consistent with an adapter that becomes more conservative about asserting object presence.

Fourth, class balance is central to the validity of the experiment. The class-collapse ablation demonstrates that a model may optimize a causal objective through label-distribution exploitation instead of improved visual grounding. Strict 50/50 balancing therefore acts as a safeguard against trivial response-policy learning.

Finally, the system-level solution is constrained by hardware. The 6 GB VRAM environment requires gradient checkpointing, `bfloat16` execution, explicit CUDA memory management, and PEFT-only checkpoint extraction. These engineering decisions are necessary to make the proposed adaptation method operational on commodity laptop hardware.

XIV. Conclusion

This work presents a complete diagnostic and mitigation pipeline for object hallucination in Vision-Language Models using the POPE benchmark.

The final system uses `SmolVLM-500M-Instruct` with LoRA adapters applied to the attention `q_proj` and `v_proj` modules. With rank

$$\boxed{r = 16}$$

and scaling factor

$$\boxed{\alpha = 32},$$

only 0.4372% of the full model parameters are trained.

The principal result is:

$$\boxed{\text{FPR} : 0.20 \rightarrow 0.16}$$

which corresponds to a

$$\boxed{20\%}$$

relative reduction in false-positive object hallucination.

At the same time, overall accuracy remains

$$\boxed{\text{Accuracy} = 0.81},$$

and the Yes/No response ratio remains near the balanced operating point:

$$\boxed{\text{Yes/No Ratio} = 0.47}.$$

The strict 50/50 training balance is a critical methodological requirement. Without it, the model exhibited class collapse, causing FPR to increase to 0.42 and affirmative bias to approximately 0.70.

Under the 6 GB VRAM constraint, gradient checkpointing and PEFT-specific checkpoint management are necessary for reliable training. The `get_peft_model_state_dict` and `set_peft_model_state_dict` workflow prevents state-dictionary collisions associated with quantized model internals.

Overall, the experiments indicate that parameter-efficient attention adaptation provides a practical mechanism for reducing object hallucination in compact Vision-Language Models. The results also show that evaluation methodology, especially class balance and bias diagnostics, is as important as the mitigation technique itself.

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